

Grade 7 Mathematics — Mock Exam — Mark schemes

PAPERS A TO D • 80 MARKS EACH • TUTOR COPY

Marks in square brackets show how the total is split. Award method marks even when the final answer is wrong. These are cross-topic papers, so each note names the unit the mark was lost in, not just the mistake — on an 80-mark script that is the thing worth reading off. Four patterns recur and are worth tracking across all four papers: dropping the sign when a negative multiplies into a bracket (2.3), reading $a : b$ as the fraction a/b (12.1), dividing a percentage change by the new value instead of the original (10.1), and treating $\div 0.1$ as $\div 10$ (3.1). He rarely makes all four; find which two are his.

Paper A — Medium

80 MARKS • 75 MINUTES

- 1** (a) $2^2 \times 3 \times 7$ [1] [5]
 (b) 42 [1] • (c) 6 [1] • (d) 81 [1] • (e) 4.5 [1]
(a) without index form ($2 \times 2 \times 3 \times 7$) is still worth the mark here — the question asks for index form, but Unit 1.1 is the skill being tested. In (d), 12 means he has multiplied 3 by 4 instead of raising it; that is a 1.4 failure, not an arithmetic slip.
- 2** (a) 6.5 [1] • (b) 0.12 [1] • (c) 0.375 [1] [5]
 (d) $3 : 4$ [1] • (e) $3x + 15$ [1]
1.2 in (b) is the standard error — counting one decimal place instead of two. $3x + 5$ in (e) means he multiplied only the first term; that same error reappears in Q16 and Q24, so check whether it is consistent before calling it careless.
- 3** (a) HCF = 6 [1] [2]
 (b) LCM = 36 [1]
Swapping the two is the whole error in 1.1. If HCF = 36 and LCM = 6 he knows both methods and has attached the wrong name to each — one mark, and a five-minute conversation, not a re-teach.
- 4** $2^{5+3-6} = 2^2$ [1] [2]
 $= 4$ [1]
 2^2 left as the final answer still earns both marks only if the question asked for index form — it did not, so the second mark needs the 4. Getting 2^{15} means he multiplied the indices instead of adding them.
- 5** (a) 62 [1] [2]
 (b) 0.062 [1]
This is 3.1 in one line: dividing by 0.1 makes a number ten times bigger. If (a) is 0.62 he has treated $\div 0.1$ as $\div 10$, which is the single most common error in Unit 3.
- 6** Writing all four to 3 d.p.: 0.680, 0.700, 0.702, 0.710 [1] [2]
0.68, 0.7, 0.702, 0.71 [1]
Putting 0.68 last means he is reading longer as larger. Award the first mark if the digits are lined up correctly even when the final order is wrong.
- 7** Common denominator 24: $\frac{16}{24}, \frac{15}{24}, \frac{14}{24}, \frac{18}{24}$ [1] [2]
 $\frac{7}{12}, \frac{5}{8}, \frac{2}{3}, \frac{3}{4}$ [1]
The answer must be given in the original fractions, not the twenty-fourths. $\frac{5}{8}$ and $\frac{7}{12}$ are the close pair; getting those two the wrong way round is a 7.2 error, while a wrong common denominator is a 1.1 error. They need different fixes.

- 8** $\frac{17}{6} - \frac{11}{6} = \frac{51}{12} - \frac{22}{12}$ [1] [2]
 $= \frac{29}{12} = 2\frac{5}{12}$ [1]
A quarter minus five sixths is negative, so this needs a borrow. An answer of $3\frac{7}{12}$ means he subtracted the smaller fraction from the larger regardless of which was on top — the defining 7.3 error.
- 9** $4p + 2(p + 3)$ [1] [2]
 $= 6p + 6$ [1]
The unsimplified form earns the first mark on its own. $4p + 2p + 3$ means he multiplied only the p inside the bracket — the same 2.3 error as Q2(e).
- 10** (a) $3(2x + 5)$ [1] [2]
 (b) $4y(2y - 3)$ [1]
In (b), $4(2y^2 - 3y)$ is not fully factorised — no mark. Taking out only y or only 4 is the 2.4 error worth naming out loud.
- 11** (a) 24 [1] [3]
 (b) -10 [1]
 (c) -6 [1]
These three sit together on purpose. If (a) is -24 and (b) is -10 he is applying the multiplication rule to a subtraction, or has not learned it at all. Two right and one wrong is carelessness; (a) wrong is a 1.2 gap.
- 12** Multiply both by 100: $468 \div 12$ [1] [3]
 $468 \div 12 = 39$ [1]
 $= 39$ [1]
Any correct equivalent-fraction method earns the first mark. An answer near 0.39 means he moved the point in only one of the two numbers, which is the 4.3 error.
- 13** $6 \times \frac{11}{4}$ [1] [3]
 $= \frac{66}{4} = \frac{33}{2}$ [1]
 $= 16\frac{1}{2}$ [1]
Partitioning $(6 \times 2) + (6 \times \frac{3}{4}) = 12 + 4\frac{1}{2}$ is equally valid and reaches the same place. An answer of $12\frac{3}{4}$ means he multiplied the whole number and left the fraction alone — the 7.4 error.
- 14** $8 \div \frac{2}{5} = 8 \times \frac{5}{2}$ [1] [3]
 $= \frac{40}{2}$ [1]
 $= 20$ [1]
An answer of $\frac{16}{5}$ means he multiplied by $\frac{2}{5}$ instead of inverting. The sanity check — dividing by something smaller than 1 must give an answer bigger than 8 — is worth more to him than the rule itself.
- 15** (a) $C = 40 \times 12 + 250$ [1] [3]
 $= 730$ [1]
 (b) $40n = 800$, so $n = 20$ [1]
(b) is 2.2 run backwards and is where the marks go. Dividing 1050 by 40 without subtracting the 250 first gives 26.25 — a non-integer answer he should have questioned.
- 16** $10x - 15$ [1] [3]
 $-2x + 8$ [1]
 $= 8x - 7$ [1]
 -2×-4 is $+8$. An answer of $8x - 23$ is the sign error and is worth two of the three marks. This is the single most repeated 2.3 error across all four papers.

- 17** $5x - 3x = 19 - 7$ [1] [3]
 $2x = 12$ [1]
 $x = 6$ [1]
Any correct route earns the marks. Moving a term without changing its sign gives $x = 13$ or $x = 3.25$; both are 2.5 errors, and the check (both sides equal 37) would have caught them.
- 18** 15% of 840 = 126 [1] [3]
 $840 + 126$ [1]
= ₹966 [1]
 840×1.15 in one line earns all three. An answer of 126 is the 10.1 error — he has found the increase and stopped, which the word “new price” asked him not to.
- 19** (a) $3 + 5 = 8$ parts, $3600 \div 8 = 450$ [1] [5]
Asha ₹1350, Bela ₹2250 [1]
(b) 2250×1.2 [1] = ₹2700 [1]
(c) $2700 - 1350 = ₹1350$ [1]
Dividing 3600 by 2 instead of by 8 is the 12.2 error. Note that (c) comes out at exactly 1350, the same as Asha’s share — a coincidence, not a method. Do not let him treat it as a check.
- 20** (a) $4n - 7 = n + 11$ [2] [5]
(b) $3n = 18$ [1] $\cdot n = 6$ [1]
Check: $4 \times 6 - 7 = 17$ and $6 + 11 = 17$ [1]
The check is a mark in its own right and he will skip it. In (a), $4n - 7 = 11$ misses the phrase “the number he first thought of” — a 2.1 reading error, not algebra. Award one of the two marks.
- 21** (a) Shop A: $96 \div 1.5 = ₹64$ per kg [2] [5]
Shop B: $155 \div 2.5 = ₹62$ per kg [1]
(b) **Shop B** [1]
(c) $6 \times 62 = ₹372$ [1]
Comparing kilograms per rupee instead is valid and reaches the same answer — do not penalise it. Two correct prices with no sentence naming the shop drops the (b) mark, which is the 12.3 habit worth drilling.
- 22** (a) Multiples of 12: 12, 24, 36 $\cdot$ of 18: 18, 36 [2] [5]
LCM = **36 seconds** [1]
(b) **9:00:36** [1]
(c) $360 \div 36 = 10$ times [1]
This is Q3 wearing a costume — the same LCM, now without the word. If he found 36 in Q3 and not here, the gap is 12.3 comprehension, not 1.1. Using $12 \times 18 = 216$ gives a time that is technically a common multiple but not the lowest; one mark.
- 23** (a) 800×1.25 [1] = ₹1000 [1] [5]
(b) 1000×0.8 [1] = ₹800 [1]
(c) **No change (0%)** [1]
The point of the question is that it lands exactly back on 800 — a 25% rise is undone by a 20% fall, not a 25% one. If he answers 5% decrease in (c) he has subtracted the percentages, which is the 10.2 error. Make him check (b) against his own answer.
- 24** (a) $3(x + 4)$ [1] = $3x + 12$ cm² [1] [5]
(b) $2(x + 4) + 2(3)$ [1] = $2x + 14$ cm [1]
(c) **$2(x + 7)$** [1]
Three units in one question: 2.2 to build it, 2.3 to expand it, 2.4 to put the bracket back. Forgetting that a perimeter has two lengths and two widths gives $x + 7$, which still factorises to nothing — if (c) is blank, check (b) first.

- 1** (a) $2 \times 3^2 \times 5$ [1] [5]
 (b) 40 [1] · (c) 14 [1] · (d) 5^2 [1] · (e) 73 [1]
(d) asks for a single power, so 25 alone does not earn it — the 1.4 skill is subtracting the indices, not evaluating. (e) is the same $\div 0.1$ test as Paper A Q5; compare the two scripts.
- 2** (a) $35\,000$ [1] · (b) 0.03 [1] · (c) 0.5 [1] [5]
 (d) $5 : 7$ [1] · (e) $7(x - 3)$ [1]
(a) 34 000 means he rounded on the wrong digit. In (c) the dot must sit over the 5; 0.56 or 0.6 is a 7.1 error. (d) needs 7 spotted as the common factor — $35 : 49$ resists the obvious 5 and 2.
- 3** (a) $\text{HCF} = 10$ [1] [2]
 (b) $\text{LCM} = 60$ [1]
20 and 30 both being multiples of 10 makes this the easy case. If he misses it here he will miss Q22 as well, where the same 1.1 idea is buried in a floor-tiling problem.
- 4** $(3^2)^3 = 3^6$, so $3^6 \div 3^4 = 3^2$ [1] [2]
 $= 9$ [1]
 3^5 means he added the indices instead of multiplying them for the outer power — the 1.4 error that separates a power of a power from a product of powers.
- 5** (a) 48 [1] [2]
 (b) 3.5 [1]
Dividing by 0.01 multiplies by 100; multiplying by 0.01 divides by 100. If both answers move the point the same way he has the direction rule, not the operation rule.
- 6** All to 4 d.p.: 0.5000 , 0.5005 , 0.5050 , 0.5500 [1] [2]
 0.5 , 0.5005 , 0.505 , 0.55 [1]
 0.5005 and 0.505 are the trap. Ranking by how many digits each has puts them in exactly the wrong order, which is the 4.1 error this set of numbers was chosen to catch.
- 7** Common denominator 20: $\frac{12}{20}$, $\frac{14}{20}$, $\frac{11}{20}$, $\frac{13}{20}$ [1] [2]
 $\frac{11}{20}$, $\frac{3}{5}$, $\frac{13}{20}$, $\frac{7}{10}$ [1]
Two of the four are already in twentieths, so this is a gentler 7.2 than Paper A Q7. Answering in twentieths instead of the original fractions costs the second mark.
- 8** $\frac{16}{3} - \frac{11}{4} = \frac{64}{12} - \frac{33}{12}$ [1] [2]
 $= \frac{31}{12} = 2\frac{7}{12}$ [1]
Needs a borrow, like Paper A Q8. $\frac{35}{12}$ is the flipped-subtraction answer. If he gets one of the two papers right and not the other, it is confidence, not method.
- 9** $3n + 4(n - 5)$ [1] [2]
 $= 7n - 20$ [1]
"5 less than" is $n - 5$, not $5 - n$. Reversing it gives $20 - n$ and is a 2.1 reading error — mark it as comprehension, not algebra.
- 10** $8x + 12 + 3x - 15$ [1] [2]
 $= 11x - 3$ [1]
Both brackets are added here, so there is no sign trap — this is the control question for Q16 on the same paper, which does have one. Getting this right and Q16 wrong isolates the sign rule.

- 11** (a) 45 **[1]** [3]
 (b) -14 **[1]**
 (c) -8 **[1]**
Same trio as Paper A Q11 with different numbers. (c) has one negative, so the answer is negative; if he writes 8 he has applied "two negatives make a positive" to a single one.
- 12** Multiply both by 100: $576 \div 24$ **[1]** [3]
 $576 \div 24 = 24$ **[1]**
 $= 24$ **[1]**
The answer happening to equal the 24 in the divisor is a coincidence worth pointing out before he decides it is a pattern.
- 13** $8 \times \frac{17}{5}$ **[1]** [3]
 $= \frac{136}{5}$ **[1]**
 $= 27\frac{1}{5}$ **[1]**
Partitioning to $24 + 3\frac{1}{5}$ is equally valid. Leaving $\frac{136}{5}$ as the final answer costs the last mark only because the question asked for a mixed number.
- 14** $12 \div \frac{3}{4} = 12 \times \frac{4}{3}$ **[1]** [3]
 $= \frac{48}{3}$ **[1]**
 $= 16$ **[1]**
An answer of 9 means he multiplied by $\frac{3}{4}$. The 7.5 check is that the answer must be bigger than 12, and it takes two seconds.
- 15** (a) $3(7) + 2(-4) = 21 - 8$ **[1]** [3]
 $= 13$ **[1]**
 (b) $3a + 8 = 26$, so $a = 6$ **[1]**
(a) crosses 2.2 with $1.2 - 2 \times -4$ is -8, so the terms subtract. An answer of 29 means he added. (b) is the formula run backwards and is the harder mark.
- 16** $6x - 12$ **[1]** [3]
 $-6x + 15$ **[1]**
 $= 3$ **[1]**
The x terms cancel completely and the answer is a number. He will distrust that and go looking for a mistake. An answer of $3 - 12x$ means -3×-5 was taken as -15 — the 2.3 sign error again.
- 17** $3x + 12 = 5x - 2$ **[1]** [3]
 $14 = 2x$ **[1]**
 $x = 7$ **[1]**
Expanding the bracket first is the only reliable route. Collecting x on the right avoids a negative coefficient — worth showing him, because he will otherwise fight $-2x = -14$.
- 18** 35% of 640 = 224 **[1]** [3]
 $640 - 224$ **[1]**
 $= ₹416$ **[1]**
 640×0.65 in one line earns all three and is the 10.2 method. An answer of 224 is the decrease, not the new price — the same stop-too-early error as Paper A Q18.
- 19** (a) $2 + 5 = 7$ parts, $4200 \div 7 = 600$ **[1]** [5]
 Kiran **₹1200**, Leela **₹3000** **[1]**
 (b) 1200×0.75 **[1]** = **₹900** **[1]**
 (c) $3000 - 900 = ₹2100$ **[1]**
Reading 2 : 5 as "two fifths to Kiran" gives 1680 and is the defining 12.1 error. It reappears in Paper D Q24(a) as a statement to diagnose.

- 20** (a) $x + (3x - 8) = 64$ [2] [5]
 (b) $4x = 72$ [1] · $x = 18$ [1]
 (c) $3(18) - 8 = 46$ [1]
"8 fewer than three times" is $3x - 8$, not $3(x - 8)$. The second form gives $x = 22$ and a brother with 42 — which still sums to 64, so the check does not catch it. That is why (a) carries two marks.
- 21** (a) Shop A: $245 \div 2.5 = ₹98$ per litre [2] [5]
 Shop B: $380 \div 4 = ₹95$ per litre [1]
 (b) **Shop B** [1]
 (c) $10 \times 95 = ₹950$ [1]
Shop A looks cheaper because the price is smaller. That is the trap and it is 12.3, not arithmetic. Dividing 245 by 2.5 also tests 4.3 — if that division is where it broke, the ratio work is fine.
- 22** (a) The tile must divide both 60 and 84, so find the HCF [1] [5]
 $60 = 2^2 \times 3 \times 5$, $84 = 2^2 \times 3 \times 7$ [1]
 HCF = **12 cm** [1]
 (b) $(60 \div 12) \times (84 \div 12) = 5 \times 7$ [1] = **35 tiles** [1]
The word HCF never appears, which is the whole test. Using the LCM gives 420 cm — a tile larger than the floor, and he should notice. In (b), $5 + 7 = 12$ instead of 5×7 is an area-versus-perimeter confusion worth naming.
- 23** (a) $1500 \times 1.2 = ₹1800$ [1] [5]
 (b) 1800×0.8 [1] = **₹1440** [1]
 (c) $1500 - 1440 = 60$, and $60 \div 1500$ [1] = **4% decrease** [1]
A 20% rise followed by a 20% fall loses 4%, because the fall is taken off a bigger number. Answering "no change" in (c) is the 10.2 error, and it is the exact opposite of the trap in Paper A Q23 — where it genuinely was no change. Check he has a reason, not a habit.
- 24** (a) $5(2x - 1)$ [1] = **$10x - 5$ cm²** [1] [5]
 (b) $2(2x - 1) + 2(5)$ [1] = **$4x + 8$ cm** [1]
 (c) **$4(x + 2)$** [1]
 $-1 \times 2 = -2$, then $+10$ gives $+8$. An answer of $4x + 12$ means he forgot to double the -1 . In (c), $2(2x + 4)$ is not fully factorised — the 2.4 error.

Paper C — Hard

80 MARKS · 75 MINUTES

- 1** (a) 37 [1] · (b) -24 [1] · (c) 11 [1] [5]
 (d) 2^6 [1] · (e) 6 [1]
(a) 39 is not prime (3×13) and is the answer he will reach by looking for "odd and near 40". (b) has three negatives, so the answer is negative — two negatives cancelling and one being left over is the 1.2 idea. (d) needs $4 = 2^2$ spotted first.
- 2** (a) 0.05 [1] · (b) 0.002 [1] · (c) 0.63 [1] [5]
 (d) $1 : 2$ [1] · (e) $6x(2x - 3)$ [1]
(a) 0.04 means he truncated instead of rounding, and 0.0498 was chosen because it rounds up twice over. In (c) both digits recur, so two dots. (d) needs the units converted first — $750 : 1.5$ is the 12.1 error and gives $500 : 1$.
- 3** (a) HCF = 12 [1] [2]
 (b) LCM = 72 [1]
24 and 36 share 12, and 72 is only twice 36 — both answers are close enough to the inputs that a guess looks plausible. Insist on the prime factorisation being visible.

- 4** $2^8 \div 2^6 = 2^2$ [1] [2]
= 4 [1]
Two indices to add on top and two on the bottom before anything is subtracted. Working the brackets out as numbers ($256 \div 64$) is valid and earns both marks, but he should be nudged to the 1.4 route.
- 5** (a) 360 [1] [2]
 (b) 0.007 [1]
(b) multiplying by 0.1 makes it smaller, and 0.07 is already small — an answer of 0.7 is the direction error. Together these two are the whole of 3.1.
- 6** All to 4 d.p.: 0.4100, 0.4010, 0.4000, 0.0409 [1] [2]
0.41, 0.401, 0.4, 0.0409 [1]
Descending, not ascending — he will answer the question he expected. 0.0409 has the most digits and is the smallest by a long way; if he puts it first he has not looked at the leading zero.
- 7** Common denominator 36: $\frac{30}{36}, \frac{28}{36}, \frac{33}{36}, \frac{26}{36}$ [1] [2]
 $\frac{13}{18}, \frac{7}{9}, \frac{5}{6}, \frac{11}{12}$ [1]
All four are close, and 36 is the only denominator that works for all of them. Using 216 (the product) is slower but correct — full marks, and a word about it afterwards.
- 8** $\frac{37}{6} - \frac{23}{8} = \frac{148}{24} - \frac{69}{24}$ [1] [2]
 $= \frac{79}{24} = 3\frac{7}{24}$ [1]
Denominators 6 and 8 need 24, not 48 — using 48 still works and costs nothing but time. The 7.3 borrow is unavoidable here because $\frac{1}{6} < \frac{7}{8}$.
- 9** Length = $w + 3$, so perimeter = $2w + 2(w + 3)$ [1] [2]
= $4w + 6$ [1]
No diagram is given, which is the difficulty. An answer of $2w + 3$ means he added a length and a width and stopped. $4w + 3$ means he doubled the bracket but not the 3.
- 10** $6x - 15 - 12 + 4x$ [1] [2]
= $10x - 27$ [1]
 $-4(3 - x) = -12 + 4x$. The x is second inside the bracket and its sign flips, which is the hardest form of the 2.3 error. $2x - 27$ is the expected wrong answer.
- 11** (a) 21 [1] [3]
 (b) -4 [1]
 (c) 15 [1]
(b) subtracting a negative adds. -10 means he treated $-(-3)$ as -3 . This is the 1.2 case the other three papers do not test.
- 12** Multiply both by 1000: $918 \div 27$ [1] [3]
 $918 \div 27 = 34$ [1]
= 34 [1]
Three decimal places to shift in both numbers. Shifting by different amounts is the 4.3 error and gives 3.4 or 340 — both look plausible, which is the point of choosing these numbers.
- 13** $15 \times \frac{14}{5}$ [1] [3]
 $= \frac{210}{5}$ [1]
= 42 [1]
Cancelling the 15 against the 5 first gives $3 \times 14 = 42$ in one step. The answer is a whole number, which he will not expect from a mixed number and may talk himself out of.

- 14** $9 \div \frac{3}{8} = 9 \times \frac{8}{3}$ [1]
 $= \frac{72}{3}$ [1]
 $= 24$ [1] [3]
Another whole-number answer. If he gets Q13 and Q14 right on this paper but not the equivalents on Paper A, it is because these cancel cleanly — not because 7.4 and 7.5 are secure.
- 15** (a) $2(9 + 6.5) = 2(15.5) = 31$ [1] [3]
 (b) $2(13 + w) = 46$, so $13 + w = 23$ [1]
 $w = 10$ [1]
(b) is 2.2 backwards. Dividing 46 by 2 first is the clean route; expanding to $26 + 2w = 46$ is equally fine. Subtracting 13 from 46 before halving gives 16.5 and is the error to watch for.
- 16** $12x + 20 - 4x + 10$ [1] [3]
 $= 8x + 30$ [1]
 $= 2(4x + 15)$ [1]
Two units in one question: expand, then factorise what comes out. $-2 \times -5 = +10$. If he reaches $8x + 10$ he will factorise to $2(4x + 5)$ and lose only the middle mark — award the third for a correct factorisation of his own wrong expression.
- 17** $6x - 2 = 4x + 12$ [1] [3]
 $2x = 14$ [1]
 $x = 7$ [1]
Brackets on both sides. Dividing through by 2 at the start gives $3x - 1 = 2(x + 3)$ and is quicker — worth showing him, but not required.
- 18** 18% of 1250 = 225 [1] [3]
 $1250 - 225$ [1]
 $= ₹1025$ [1]
 1250×0.82 earns all three. 18% is awkward without a calculator: $10\% = 125$, $8\% = 100$. If he reaches for long multiplication instead of partitioning, that is the 10.2 habit to fix.
- 19** (a) $2 + 3 + 4 = 9$ parts, $9450 \div 9 = 1050$ [1] [5]
 $₹2100$, $₹3150$ [1] and $₹4200$ [1]
 (b) $4200 \times 1.12 = ₹4704$ [1]
 (c) $2100 + 3150 + 4704 = ₹9954$ [1]
(c) is where it goes wrong: the other two shares are unchanged, so only the largest is increased. Adding 12% to the whole 9450 gives 10 584 and is the error. Check the three shares in (a) sum to 9450 before anything else.
- 20** (a) $x + (3x - 20) + (x + 40) = 180$ [1] [5]
 $5x + 20 = 180$ [1] · $x = 32$ [1]
 (b) Angles are 32° , 76° and 72° [1] · largest = 76° [1]
The 180 is not given and must be recalled — that is deliberate. In (b) the largest angle is $3x - 20$, not $x + 40$, so he has to evaluate all three rather than guess from the expressions. Answering 32° means he stopped at x .
- 21** (a) $57 \div 0.75 = ₹76$ [1] · $90 \div 1.25 = ₹72$ [1] [5]
 $185 \div 2.5 = ₹74$ [1]
 (b) 1.25 litres [1]
 (c) $5(76 - 72) = ₹20$ [1]
The biggest bottle is not the best value, which is the whole design. Three divisions by decimals makes this 4.3 as much as 12.3. In (c) the worst value is the smallest bottle at 76 — using 74 gives ₹10 and means he compared against the wrong size.

- 22** (a) $15 = 3 \times 5$, $24 = 2^3 \times 3$ [1] [5]
 LCM = $2^3 \times 3 \times 5$ [1] = **120 days** [1]
 (b) 120, 240 and 360 all fall within 365 [1] = **3 times** [1]
15 × 24 = 360 is a common multiple but not the lowest, and it gives a believable 1 time in (b) — the 1.1 error this question was built around. In (b), $365 \div 120 = 3.04$, and rounding that to 3 is right for the wrong reason; ask him to name the three dates.
- 23** (a) $2000 \times 1.3 = \text{₹}2600$ [1] [5]
 (b) 2600×0.7 [1] = **₹1820** [1]
 (c) Loss = $2000 - 1820 = 180$, and $180 \div 2000$ [1] = **9% decrease** [1]
 $1.3 \times 0.7 = 0.91$, so a 30% rise and a 30% fall lose 9%. Dividing the 180 by 1820 instead of 2000 gives 9.9% and is the classic percentage-change error — the base is always the original. Award the first three marks only.
- 24** (a) $4(3x + 2)$ [1] = **$12x + 8 \text{ cm}^2$** [1] [5]
 (b) $2(3x + 2) + 2(4)$ [1] = **$6x + 12 \text{ cm}$** [1]
 (c) **$6(x + 2)$** [1]
In (c), $2(3x + 6)$ and $3(2x + 4)$ are both incomplete — “fully” means the HCF of 6 and 12, which is 6. That is 1.1 doing the work inside a 2.4 question, and it is the mark most often dropped on this paper.

Paper D — Hard

80 MARKS · 75 MINUTES

- 1** (a) 2 (they are 23 and 29) [1] [5]
 (b) 8 [1] · (c) 7 [1] · (d) 9^7 [1] · (e) 50 [1]
(a) asks how many, not which — answering “23 and 29” is still the mark. Counting 21 (3×7) or 27 (3^3) as prime is the 1.1 error. (d) 9^{10} means he multiplied the indices.
- 2** (a) 7.10 [1] · (b) 0.036 [1] · (c) 0.26 [1] [5]
 (d) $1 : 3$ [1] · (e) $5a(4a + 3)$ [1]
(a) must keep the trailing zero — 7.1 is not 2 decimal places. In (c) only the 6 recurs, so the dot goes on the 6 alone; a dot over the 2 as well is the 7.1 error. (d) needs hours converted to minutes before simplifying.
- 3** (a) HCF = 9 [1] [2]
 (b) LCM = 90 [1]
18 and 45 share 9, not 3 — stopping at 3 is the usual 1.1 slip. Neither number divides the other, which rules out the shortcut that works on Paper B Q3.
- 4** $5^{4+3-5} = 5^2$ [1] [2]
 = **25** [1]
Add the two on top, then subtract the bottom. The question asks for an ordinary number, so 5^2 alone is one mark.
- 5** (a) 0.62 [1] [2]
 (b) 6.2 [1]
The same digits appear in both answers, one hundred times apart. If he gives 0.62 twice he has read both operations as the same one — the core 3.1 confusion.
- 6** All to 4 d.p.: 0.0330, 0.3003, 0.3030, 0.3300 [1] [2]
0.033, 0.3003, 0.303, 0.33 [1]
Three of the four start 0.3 and are separated only in the third and fourth places. Lining up the decimal points on paper is the method; comparing them in his head is the 4.1 error.

- 7** Common denominator 24: $\frac{21}{24}, \frac{20}{24}, \frac{22}{24}, \frac{17}{24}$ **[1]** [2]
 $\frac{11}{12}, \frac{7}{8}, \frac{5}{6}, \frac{17}{24}$ **[1]**
Descending. $\frac{11}{12}$ and $\frac{7}{8}$ are one twenty-fourth apart, which no amount of looking will separate. Answering in ascending order costs the second mark only.
- 8** $\frac{37}{5} - \frac{29}{8} = \frac{296}{40} - \frac{145}{40}$ **[1]** [2]
 $= \frac{151}{40} = 3\frac{31}{40}$ **[1]**
Denominator 40 and an awkward numerator — the hardest 7.3 of the four papers. A whole-number part of 4 means he did $7 - 3$ and then took the fractions the easy way round.
- 9** (a) $50 + 18k$ **[1]** [2]
 (b) $2(50 + 18k) = 100 + 36k$ **[1]**
(b) $100 + 18k$ means he doubled only the fixed fee. Either the bracketed or the expanded form earns the mark in (b), but the doubling must reach both terms — 2.1 into 2.3.
- 10** $10x - 6 - 5x + 10$ **[1]** [2]
 $= 5x + 4$ **[1]**
 $-5 \times -2 = +10$. An answer of $5x - 16$ is the sign error, and it is the fourth appearance of that same 2.3 mistake across the set. If he has made it on all four papers it is a rule he does not have, not a slip.
- 11** (a) 54 **[1]** [3]
 (b) -15 **[1]**
 (c) -12 **[1]**
Same trio, fourth paper. By now the pattern across the four scripts is the useful thing: if (a) is always right and (b) sometimes wrong, the gap is in adding negatives, not multiplying them.
- 12** Multiply both by 1000: $2622 \div 38$ **[1]** [3]
 $2622 \div 38 = 69$ **[1]**
 $= 69$ **[1]**
The hardest division on the four papers, and the only one where the long division itself is real work. Award the first two marks for a correct set-up even if the arithmetic then fails — that separates 4.3 from plain calculation.
- 13** $14 \times \frac{12}{7}$ **[1]** [3]
 $= \frac{168}{7}$ **[1]**
 $= 24$ **[1]**
Cancelling 14 against 7 gives 2×12 straight away. An answer of $14\frac{5}{7}$ means he multiplied the whole number and copied the fraction across — the 7.4 error.
- 14** $10 \div \frac{2}{7} = 10 \times \frac{7}{2}$ **[1]** [3]
 $= \frac{70}{2}$ **[1]**
 $= 35$ **[1]**
Dividing by $\frac{2}{7}$ more than triples the 10. If his answer is under 10 he has inverted the wrong number or not inverted at all — 7.5, and the estimate would have caught it.
- 15** (a) $1.8(25) + 32 = 45 + 32 = 77$ **[1]** [3]
 (b) $1.8C = 5 - 32 = -27$ **[1]**
 $C = -27 \div 1.8 = -15$ **[1]**
(b) crosses three units: 2.2 to rearrange, 1.2 for the negative, 4.3 to divide by 1.8. A positive answer means he subtracted the wrong way round. 15 instead of -15 earns the first mark only.

- 16** $14x - 21$ [1] [3]
 $-8x + 24$ [1]
 $= 6x + 3$ [1]
 $-4 \times -6 = +24$, so the constant is positive. $6x - 45$ is the sign error. Unlike Paper B Q16 the x terms do not cancel, so a bare number here is wrong.
- 17** $8x - 12 = 3x + 18$ [1] [3]
 $5x = 30$ [1]
 $x = 6$ [1]
Both sides bracketed. Check: $4(12 - 3) = 36$ and $3(6 + 6) = 36$. He will not check unless told to, and this is the paper where it matters most.
- 18** Decrease = $640\,000 - 544\,000 = 96\,000$ [1] [3]
 $96\,000 \div 640\,000$ [1]
 $= 15\%$ decrease [1]
Dividing by 544 000 instead of 640 000 gives 17.6% — the most common error in Unit 10, and the reason these numbers were chosen. The base is the original value. Award the first mark only.
- 19** (a) $5 + 4 = 9$ parts, $7200 \div 9 = 800$ [1] [5]
Asha ₹4000, Bina ₹3200 [1]
(b) 15% of 4000 = ₹600 [1]
(c) Asha ₹3400 [1], Bina ₹3800 [1]
Bina ends up with more than Asha, which he will read as a mistake and may “correct”. The total is still 7200 — that is the check. Taking 15% of Bina’s share instead of Asha’s gives 480 and breaks the total.
- 20** (a) Length = $2w + 5$, so $2(w + 2w + 5) = 58$ [2] [5]
(b) $3w + 5 = 29$ [1] · $w = 8$ cm [1]
(c) Length = $2(8) + 5 = 21$ cm [1]
Halving the 58 first is the clean route. “5 more than twice” is $2w + 5$, not $2(w + 5)$ — the second form gives $w = 6$ and a 17 cm length, and $2(6 + 17) = 46$, not 58, so the check catches it. Make him do the check.
- 21** (a) $91 \div 0.35$, multiply both by 100: $9100 \div 35$ [1] [5]
 $9100 \div 35$ [1] = 260 bottles [1]
(b) 8×0.35 [1] = ₹2.80 [1]
(a) is 4.3 with a real division behind it. An answer near 31.85 means he multiplied instead of dividing — and 31 bottles from 91 litres should have looked wrong. In (b), ₹2.8 is fine; ₹28 is a 4.2 place-value error.
- 22** (a) Lowest power of each shared prime: $2^2 \times 3 \times 5$ [1] = 60 [1] [5]
(b) Highest power of each prime: $2^3 \times 3^2 \times 5^2$ [1] = 1800 [1]
(c) $n = 4 \times 3 \times 25 = 300$ [1]
The numbers are never given, only their factorisations — that is the point, and it is the hardest 1.1 question on the four papers. Swapping the rules gives HCF = 1800 and LCM = 60, which is the same name-confusion as Paper A Q3 but much harder to spot. Full working is the only way to tell.
- 23** (a) $45\,000 \times 0.8$ [1] = ₹36 000 [1] [5]
(b) $36\,000 \times 1.25$ [1] = ₹45 000 [1]
(c) No change (0%) [1]
 $0.8 \times 1.25 = 1$ exactly, so it lands back on the original price. He will assume he has made an error in (b) and go looking for it. A 20% fall is undone by a 25% rise — the mirror of Paper A Q23, and worth marking the two side by side.

- 24** (a) The mistake: reading the colon as a fraction bar. There are $3 + 5 = \mathbf{8}$ parts, not 5 [5]
[1]
The correct fraction is $\frac{3}{8}$ [1]
(b) The mistake: -2×-5 is **+10**, not -10 [1]
 $4 - 2x + 10$ [1] = **14 - 2x** [1]
These are the two errors the whole set is built around, one from Unit 12 and one from Unit 2, which is why they close the last paper. Naming the rule is worth more than writing the right answer — a student who can diagnose both in someone else's handwriting will not make them in his own, and that is the entire reason this question type exists.